

练习题

1. 交换积分次序:

$$(1) I_1 = \int_0^2 dx \int_0^{\frac{x^2}{2}} f(x, y) dy + \int_2^{2\sqrt{2}} dx \int_0^{\sqrt{8-x^2}} f(x, y) dy.$$

$$(2) I_2 = \int_0^1 dx \int_0^{x^2} f(x, y) dy + \int_1^3 dx \int_0^{\frac{1}{2}(3-x)} f(x, y) dy.$$

$$\text{解: } I_1 = \int_0^2 dy \int_{\sqrt{2y}}^{\sqrt{8-y^2}} f(x, y) dx. \quad I_2 = \int_0^1 dy \int_{\sqrt{y}}^{3-2y} f(x, y) dx.$$

2. 计算 $I_3 = \iint_D x^2 e^{-y^2} dx dy$, 其中 D 是以 $O(0, 0), A(1, 1), B(0, 1)$ 为顶点的三角形围成的闭区域.

$$\text{解: } I_3 = \int_0^1 dy \int_0^y x^2 e^{-y^2} dx = \frac{1}{3} \int_0^1 y^3 e^{-y^2} dy = -\frac{1}{6} (2e^{-1} - 1) = \frac{1}{6} (1 - \frac{2}{e}).$$

$$3. \text{ 计算 } I_4 = \int_0^a dx \int_{-x}^{-a+\sqrt{a^2-x^2}} \frac{1}{\sqrt{4a^2-x^2-y^2}} dy.$$

$$\begin{aligned} \text{解: } I_4 &= \iint_D \frac{1}{\sqrt{4a^2-r^2}} r dr d\theta = \int_{-\frac{\pi}{4}}^0 d\theta \int_0^{-2a \sin \theta} \frac{1}{\sqrt{4a^2-r^2}} r dr \\ &= \int_{-\frac{\pi}{4}}^0 (2a - 2a \cos \theta) d\theta = 2a(\theta - \sin \theta) \Big|_{-\frac{\pi}{4}}^0 = (\frac{\pi}{2} - \sqrt{2})a. \end{aligned}$$

$$4. \text{ 求 } I_5 = \iint_D \sqrt{|y-x^2|} dx dy, \text{ 其中 } D: 0 \leq x \leq 1, 0 \leq y \leq 2.$$

解: 用曲线 $y = x^2$ 将区域分成 D_1, D_2 两块, 如图

$$\begin{aligned} \text{于是 } I_5 &= \int_0^1 dx \int_{x^2}^2 \sqrt{y-x^2} dy + \int_0^1 dx \int_0^{x^2} \sqrt{x^2-y} dy = \frac{2}{3} \int_0^1 (2-x^2)^{\frac{3}{2}} dx + \\ &\frac{2}{3} \int_0^1 x^3 dx = \frac{\pi}{4} + \frac{5}{6}. \end{aligned}$$

5. 将积分 $I_6 = \int_0^1 dx \int_0^1 f(x, y) dy$ 化为极坐标系下的二次积分.

$$\text{解: } I_6 = \int_0^{\frac{\pi}{4}} d\theta \int_0^{\sec \theta} f(r \cos \theta, r \sin \theta) r dr + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} d\theta \int_0^{\csc \theta} f(r \cos \theta, r \sin \theta) r dr$$

6. 求位于两圆 $r = 2 \sin \theta, r = 4 \sin \theta$ 之间的均匀薄片的质心.

$$\text{解: } \iint_D y d\sigma = \iint_D r^2 \sin \theta dr d\theta = \int_0^{\pi} \sin \theta d\theta \int_{2 \sin \theta}^{4 \sin \theta} r^3 dr = \frac{16}{3} \int_0^{\pi} \sin^4 \theta d\theta = 7\pi.$$

于是, $y = \frac{7\pi}{3\pi} = \frac{7}{3}$, 故所求质心为 $(0, \frac{7}{3})$.